ACTIN Report (research use only)

PATIENT
EXAMPLE-LUNG-01

REPORT DATE
17-Sep-2025

Gender: Female | Birth year: 1975 | WHO: 1

Tumor: Lung adenocarcinoma | Lesions: Liver, Lung | Stage: IV

Clinical summary

Relevant systemic treatment history	**2023-06 to 2025-01 Osimertinib**

Recent molecular results

Hartwig WGS (22-Feb-2025)

Molecular tissue of origin prediction	**Lung: Non-small cell: LUAD (98%)**
Tumor mutational load / burden	**TML 160 / TMB 14 mut/Mb**
Microsatellite (in)stability	**Stable**
HR status	**Proficient (0)**
Driver mutations	**EGFR C797S, EGFR L858R, KRAS G12C, KRAS G12D**
Amplified genes	**None**
Deleted genes	**TP53 del**
Homozygously disrupted genes	**None**
Gene fusions	**MET(exon13)::MET(exon15) fusion**
Driver virus	**None**
HLA-A	**HLA-A*01:01, HLA-A*02:01**

Trial-relevant IHC results

PD-L1	**Score > 50%**

Phase 2/3+ (or unknown phase) trials in NL that are open and potentially eligible (2 trials)

TRIAL	COHORT	MOLECULAR	SITES	WARNINGS
METC 04 TEDR1 (Phase 2)	Lung cancer C797S cohort	EGFR C797S	NKI-AvL	Active CYP3A4/5 inducing medication in provided medications (St. John's Wort), At least 1 line(s) of EGFR inhibitor targeted therapy in provided treatments
EGFR-C797S-TRIAL (Phase 2)	*EGFR C797S*	*EGFR C797S*	*Elisabeth-TweeSteden Ziekenhuis*	

Trials matched solely on molecular event and tumor type (no clinical data used) are shown in italicized, smaller font.

Phase 1/2 trials in NL that are open and potentially eligible (2 trials)

TRIAL	COHORT	MOLECULAR	SITES	WARNINGS
METC 02 KAYRAS (Phase 1/2)	Dose expansion - monotherapy - NSCLC	KRAS G12D, PD-L1 >= 50.0	Erasmus MC	Variant(s) G12D in KRAS but subclonal likelihood of > 50%
METC 01 IEMOEN (Phase 1)	Dose escalation - monotherapy *(no slots)*	None		History of rheumatoid arthritis, SOC is not exhausted (at least platinum doublet remaining)

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Trials marked with asterisk (*) were sourced from ClinicalTrials.gov on 17-Sep-2025. No modifications have been made to the ClinicalTrials.gov data. ACTIN structures trial information for matching and analytical purposes.
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International trials that are open and potentially eligible (2 trials)

TRIAL	COHORT	MOLECULAR	SITES
EGFR-BE (Phase 1)	*EGFR L858R*	*EGFR L858R*	*Belgium: Brussels*
KRAS-G12C-TRIAL-DE (Phase 1)	*KRAS G12C*	*KRAS G12C*	*Germany: Stuttgart*

Trials in this table are matched solely on molecular event and tumor type (clinical data excluded).

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Molecular Details

Hartwig WGS (EXAMPLE-LUNG-01-T, 22-Feb-2025)

General

PURITY	PLOIDY	TML STATUS	TMB STATUS	MS STABILITY	HR STATUS	DPYD	UGT1A1
50%	2.3	**High (160)**	**High (14)**	Stable	Proficient (0)	*1_HOM (Normal function)	*1_HOM (Normal function)

Predicted tumor origin

	1. LUNG: NON-SMALL CELL: LUAD
Combined prediction score	**98%**
This score is calculated by combining information on:	
(1) SNV types	60%
(2) SNV genomic localisation distribution	70%
(3) Driver genes and passenger characteristics	80%

Other cohorts have a combined prediction of 2% or lower

Key drivers

TYPE	DRIVER	TRIALS (LOCATIONS)	TRIALS IN HARTWIG	BEST EVIDENCE IN EXTERNAL	RESISTANCE IN EXTERNAL
Mutation (gain of function)	EGFR L858R (2/4 copies)		NCT00000007	Approved	
Mutation (gain of function)	EGFR C797S (1/4 copies)	TEDR1 (NKI-AvL)	NCT00000008	Pre-clinical	
Mutation (gain of function)	KRAS G12D (0.3/2 copies)*	KAYRAS (Erasmus MC)			

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Hartwig WGS (EXAMPLE-LUNG-01-T, 22-Feb-2025)

Key drivers

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TYPE	DRIVER	TRIALS (LOCATIONS)	TRIALS IN HARTWIG	BEST EVIDENCE IN EXTERNAL	RESISTANCE IN EXTERNAL
Mutation (gain of function)	KRAS G12C (0.3/2 copies)*		NCT00000009		
Deletion	TP53 del				
Known fusion	MET(exon13)::MET(exon15) fusion Domain(s) kept: Tyrosine Kinase Domain(s) lost: Juxtamembrane				

* Variant has > 50% likelihood of being sub-clonal

Other drivers or relevant events

None

Immunology

HLA GENE	TYPE	TUMOR COPY NUMBER	MUTATED IN TUMOR
HLA-A	HLA-A*01:01	1.0	No
	HLA-A*02:01	0.0	No

IHC results

PD-L1	**Score > 50%**

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Clinical Details

Clinical summary

Relevant systemic treatment history	**2023-06 to 2025-01**	**Osimertinib**
Relevant other oncological history	**None**	
Previous primary tumor	**None**	
Relevant non-oncological history	**2023**	**Rheumatoid arthritis**
Toxicities grade >= 2 or unknown	**None**	
LVEF	**50%**	
Known allergies	**None**	
Recent surgeries	**None**	

Tumor details

Measurable disease	**Yes**
Lesions	**Liver, Lung**
No lesions present	**CNS, Brain, Bone, Lymph node**

Active medication details

MEDICATION	ADMINISTRATION	START DATE	STOP DATE	DOSAGE	FREQUENCY
St. John's Wort	Oral	01-Feb-2023		300 MILLIGRAMS	1 / 2 DAYS

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Trial Matching Details

Trials and cohorts that are potentially eligible, but are closed (1 trial)

TRIAL	COHORT	MOLECULAR	SITES	WARNINGS
METC 01 IEMOEN (Phase 1)	Dose expansion - monotherapy	None		History of rheumatoid arthritis, SOC is not exhausted (at least platinum doublet remaining)

Trials and cohorts that are considered ineligible (4 cohorts from 3 trials)

TRIAL	COHORT	MOLECULAR	INELIGIBILITY REASONS
METC 02 KAYRAS (Phase 1/2)	Dose expansion - monotherapy - Colorectum	KRAS G12D, PD-L1 >= 50.0	Liver metastases in provided lesions, No colorectal cancer
METC 03 NO-SEE797ES	Dose escalation - monotherapy	EGFR C797S	C797S in EGFR in canonical transcript
METC 05 PICKME3CA	*Applies to all cohorts below*	None	No PIK3CA activating mutation(s)
	Dose expansion - monotherapy - NSCLC *(closed)*		
	Dose expansion - monotherapy - Other cancer types *(closed)*		Tumor belongs to DOID term(s) lung non-small cell carcinoma

Trials and cohorts that are not evaluable (0 trials)

None

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